AI Design of Quantum Processors

Components 1–3 — from classical and quantum data to a trained classical→quantum model

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CONTEXT

We test whether classical data can predict quantum properties — first on the exactly-solvable oscillator, now on a real fluxonium qubit.

Component 1

Classical data

Phase-space orbits & energies — the model INPUTS

→

Component 2

Quantum data

Spectra, dynamics & Wigner functions — the model TARGETS

→

Component 3

Machine learning

Trained: predicts fluxonium quantum trajectories from classical ones

Central question: how much of a quantum system can be predicted from classical information alone — and where does that prediction break down?

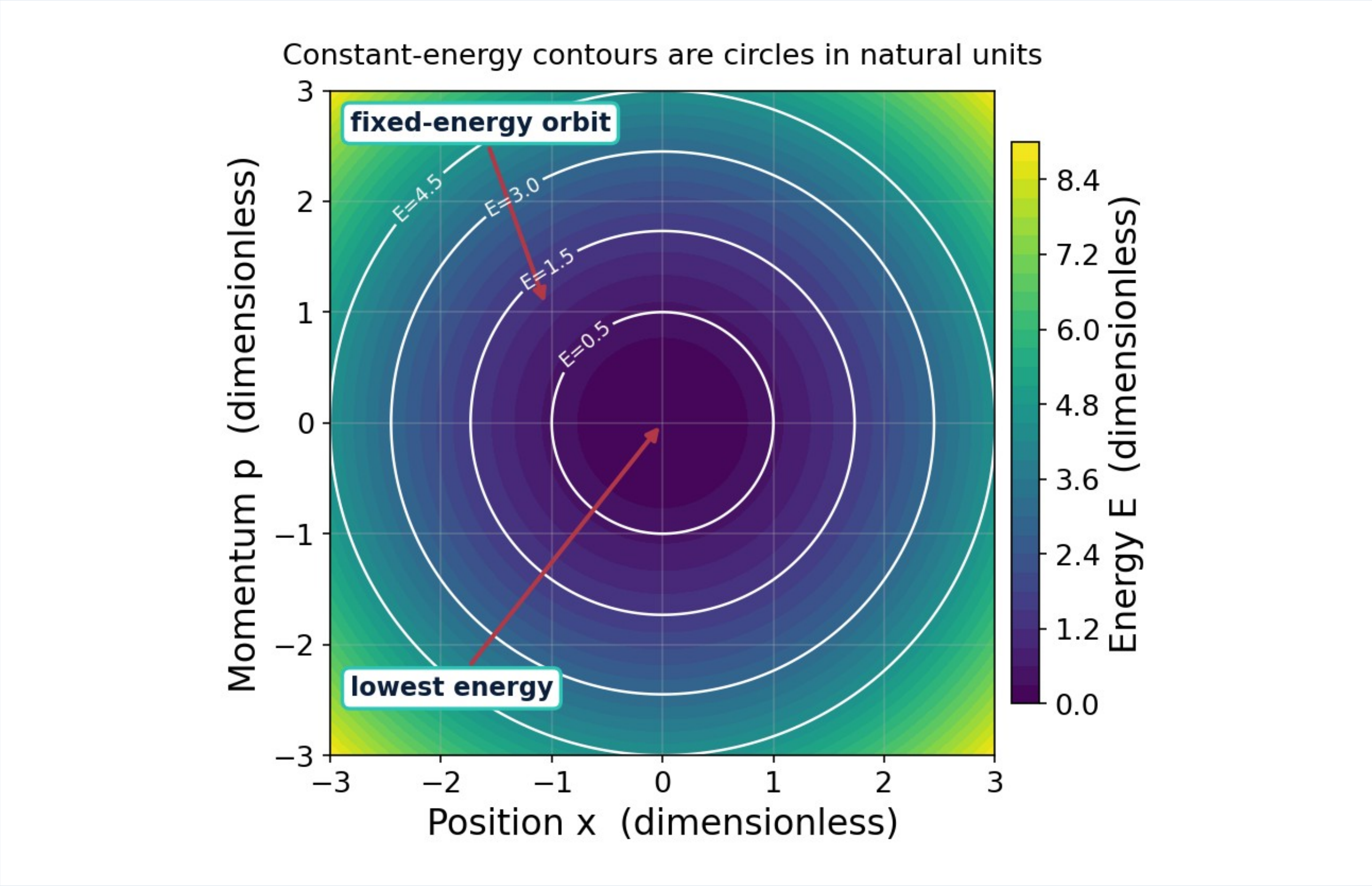
Why this system?
The harmonic oscillator is exactly solvable in both worlds — every number has a built-in answer key.

Results

Classical and quantum data, extended to the anharmonic and coupled oscillators and to the fluxonium qubit — every result checked against an exact formula.

Component 1 · Task 1

Constant-energy contours are closed elliptical orbits in phase space.

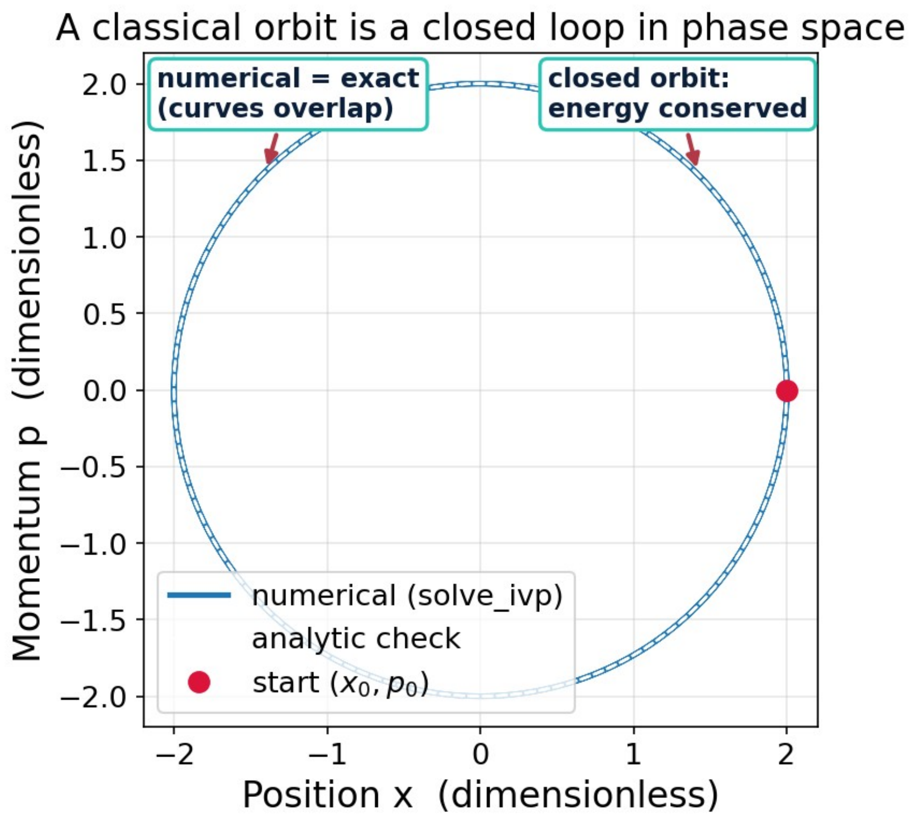


WHAT IT SHOWS

- Color is total energy E(x,p); white rings are constant-energy curves.
- Each ring is one allowed classical orbit — the system is locked onto it.
- In natural units the ellipses become circles; larger energy = larger ring.
- This map defines the classical states the ML model will draw inputs from.

Component 1 · Task 2

Numerical trajectories match the exact solution and conserve energy.

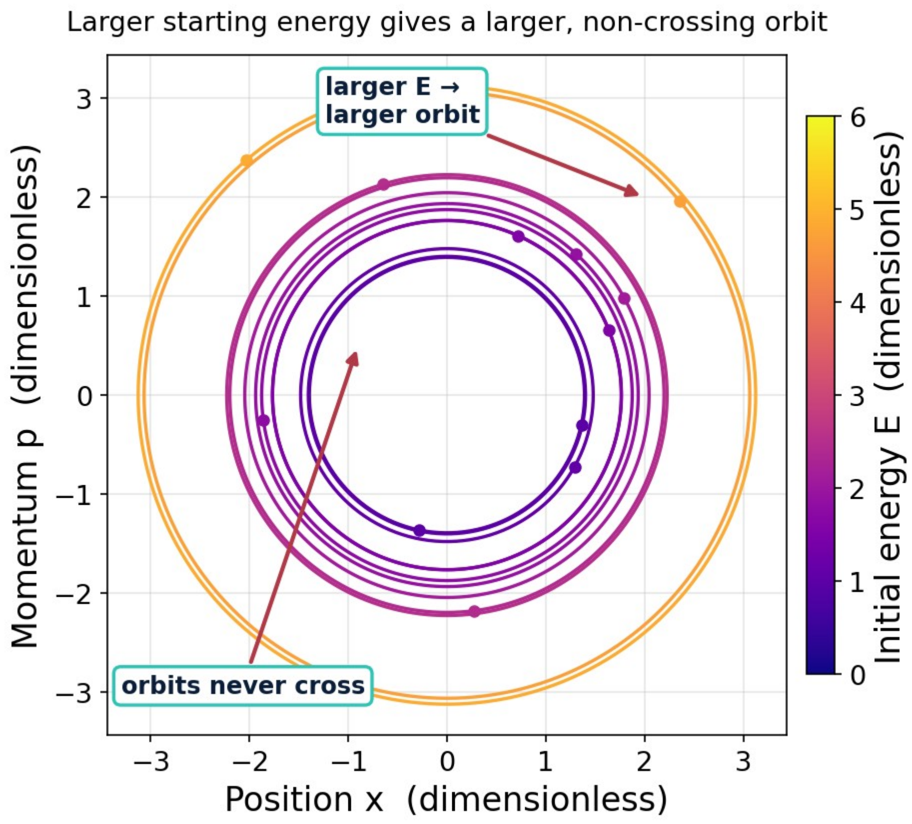


WHAT IT SHOWS

- solve_ivp steps Hamilton's equations forward in time to trace the orbit.
- Numerical path overlaps the analytic solution — the solver is verified.
- The orbit closes on itself and energy drift is only ~1e-9 — the solver conserves energy as it should.
- Confidence here transfers to systems with no exact answer later.

Component 1 · Task 2

Higher energy gives larger, non-crossing orbits — the classical feature set.

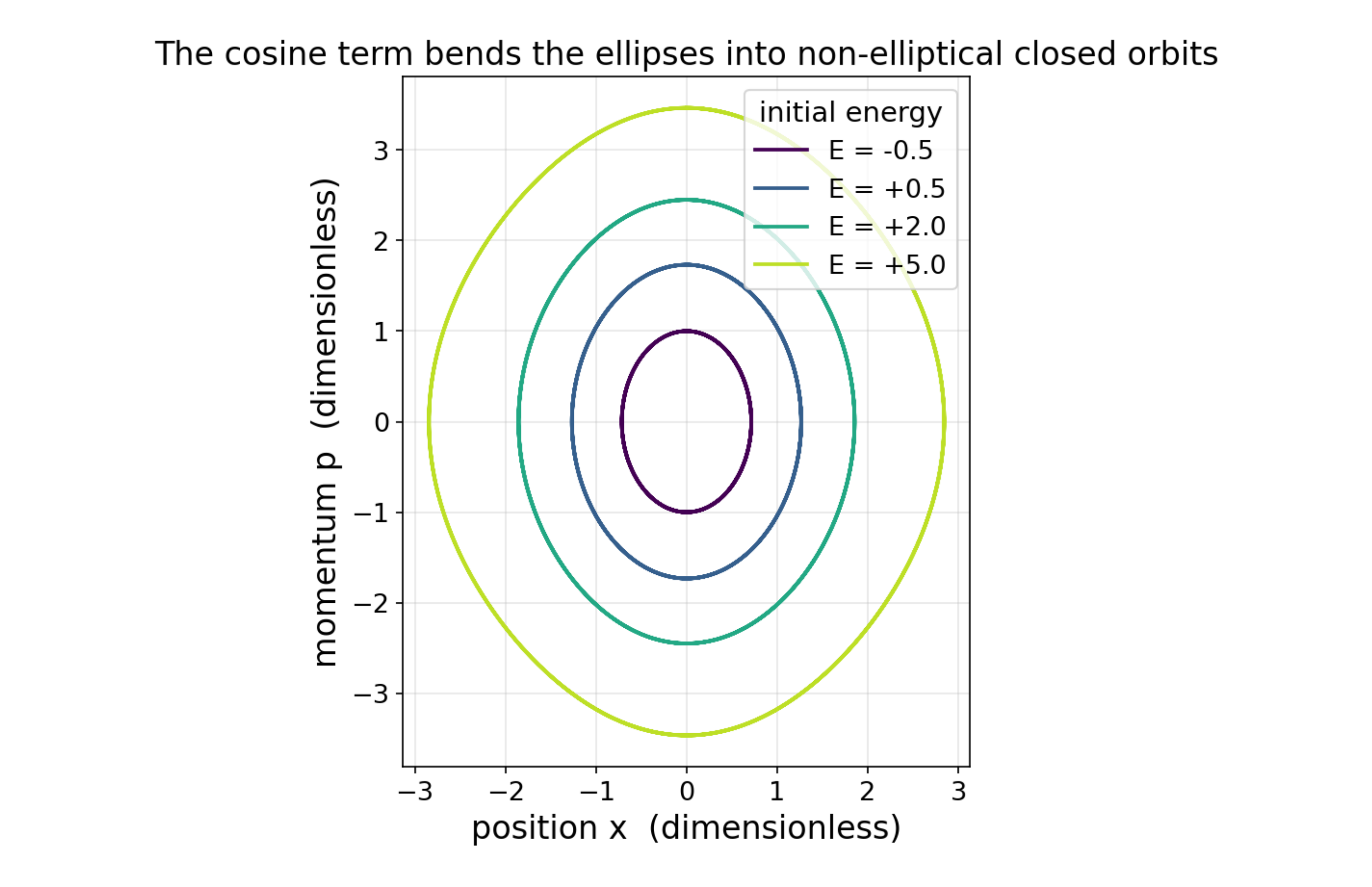


WHAT IT SHOWS

- Twelve random initial conditions, each colored by its conserved energy.
- Orbits never intersect: a unique state has a unique future.
- Saved as classical_trajectories.npy — the inputs for Component 3.
- Nested structure is exactly the 'feature space' ML will learn from.

Component 1 · Task 3

Adding a cosine well bends the orbits out of ellipses and ties the period to energy.

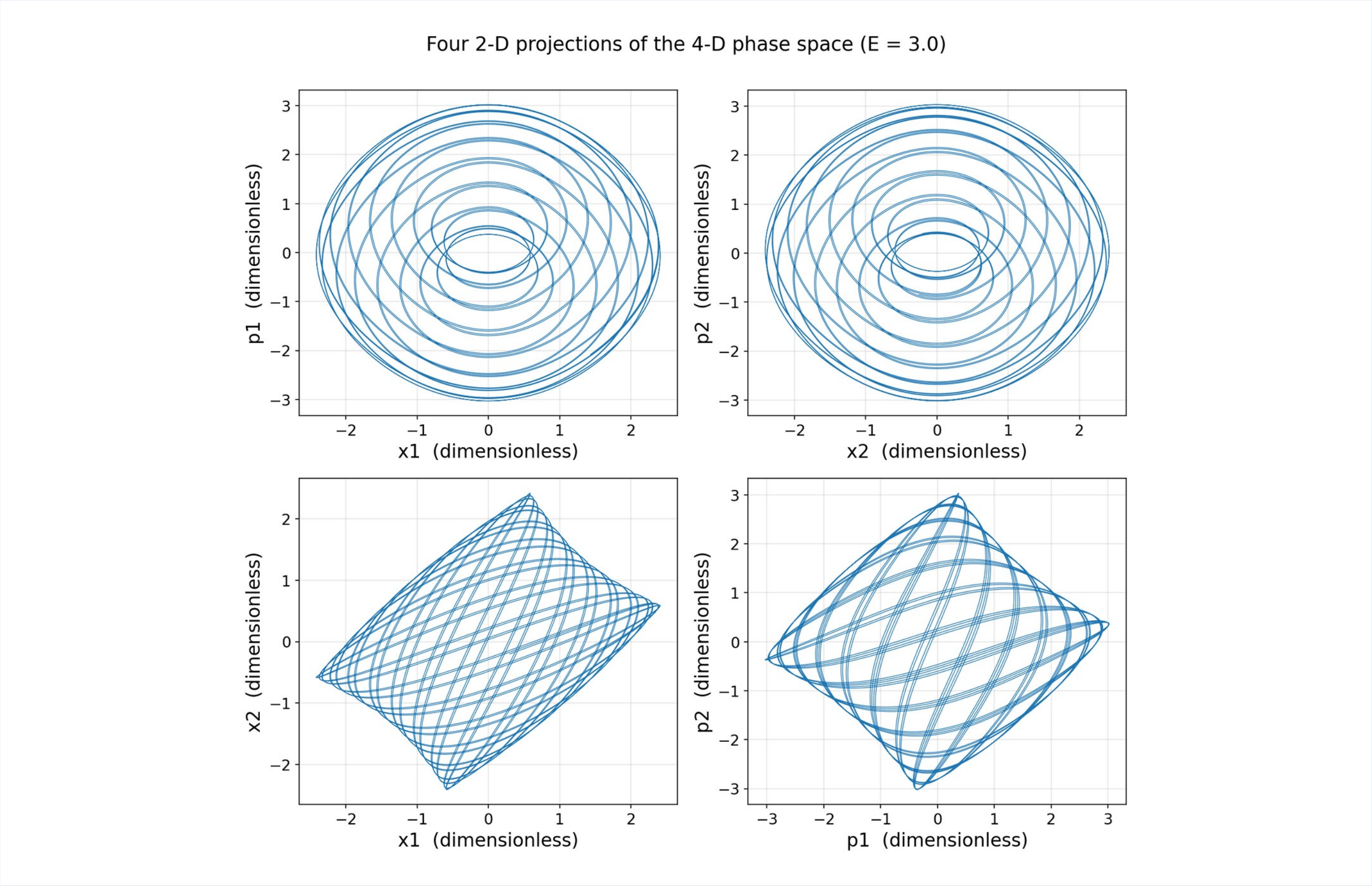


WHAT IT SHOWS

- $H = p^2/2m + \frac{1}{2}m\omega^2x^2 - V_0\cos(kx)$ with $V_0 = k = 1$ — the classical shadow of a Josephson junction.
- Low-energy orbits still look elliptical; higher-energy orbits visibly deform.
- Anharmonicity is what makes a qubit addressable — evenly spaced levels cannot be driven two at a time.
- These are the classical inputs Component 3 pairs with fluxonium targets.

Component 1 · Task 4

Two coupled oscillators live in a 4-D phase space, so we read them through 2-D projections.

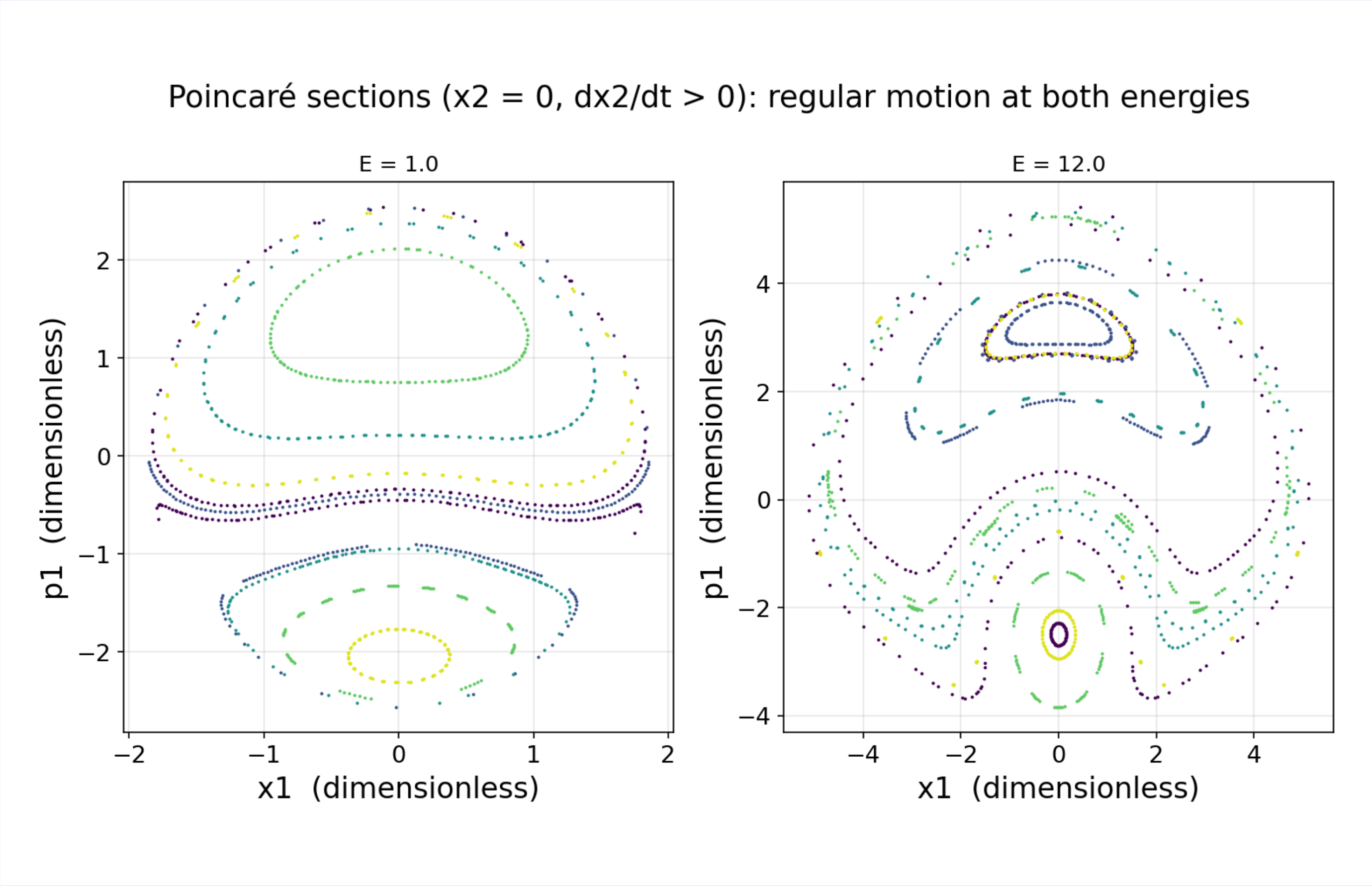


WHAT IT SHOWS

- Two cosine oscillators joined by momentum coupling $\lambda p_1 p_2$ with $\lambda = 0.3$.
- The state is (x_1, p_1, x_2, p_2) — four numbers, so the motion cannot be drawn directly.
- Each panel projects that 4-D motion onto one pair of coordinates.
- Coupling lets the two trade energy: neither conserves its own any more, only the total does.

Component 1 · Task 4

I expected chaos at high energy. Measuring it properly says the motion stays regular.

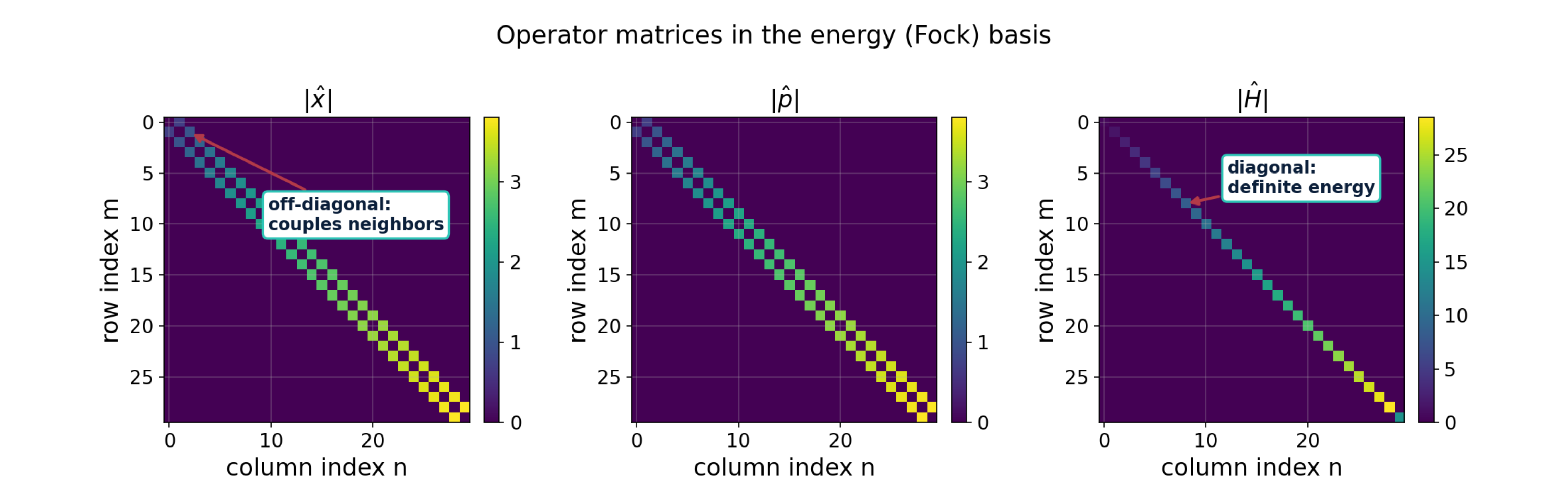


WHAT IT SHOWS

- A Poincaré section records the state only as it crosses $x_2 = 0$ in one direction — a tangled orbit becomes dots.
- Smooth nested curves mean regular, quasiperiodic motion (KAM tori); a scattered cloud would mean chaos.
- Both E = 1 and E = 12 give nested curves. The Lyapunov exponent confirms it: $\lambda \approx 0.004$–0.007, zero to within the log t/t floor.
- The nonlinearity is a bounded cosine while the harmonic term grows without limit — so higher energy makes this system MORE nearly integrable.

Component 2 · Task 1

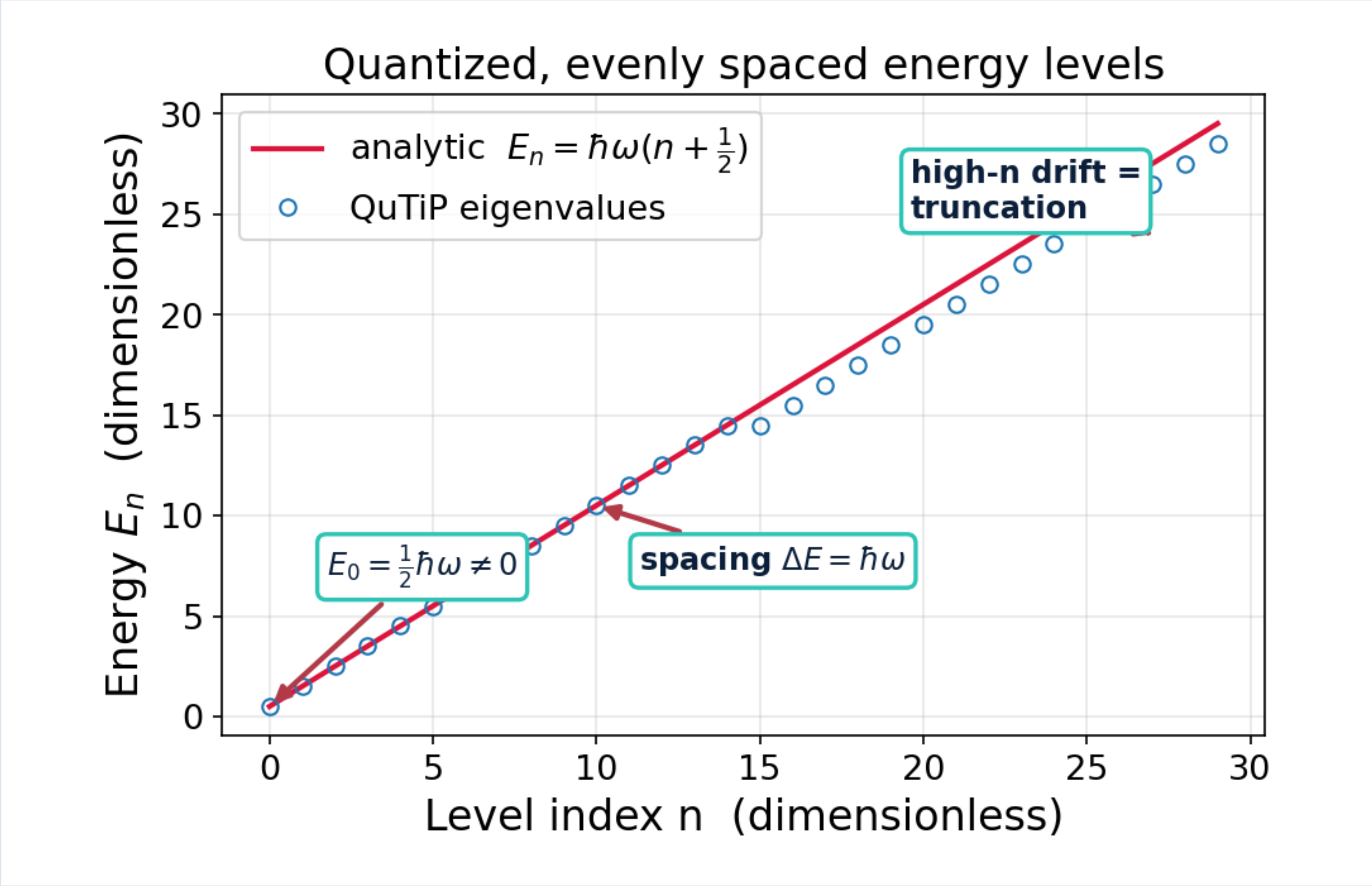
The quantum operators reveal the ladder structure of the oscillator.



Takeaway: H is diagonal (each level has a definite energy), while x̂ and p̂ have only off-diagonal entries — they connect a level to its immediate neighbors.

Component 2 · Task 1

Quantum energy is discrete, evenly spaced, with a nonzero zero-point energy.

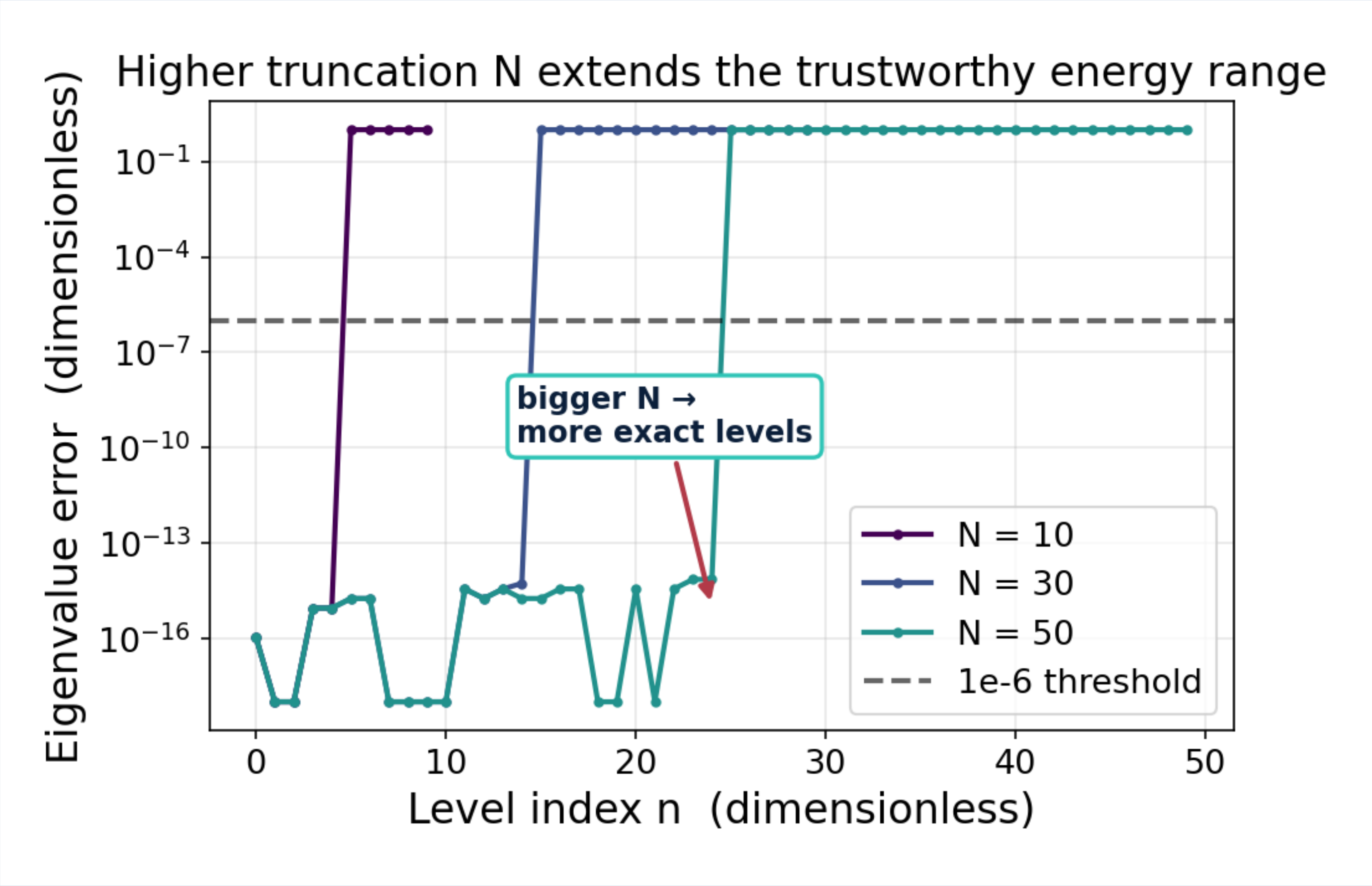


WHAT IT SHOWS

- QuTiP eigenvalues land exactly on the analytic line $E_n = \hbar\omega(n+½)$.
- Ground state $E_0 = ½\hbar\omega \neq 0$ — a purely quantum effect.
- Levels are equally spaced by $\hbar\omega$ (the ladder).
- High-n points drift: the expected signature of finite truncation.

Component 2 · Task 1

The truncation level N sets how many energy levels are trustworthy.

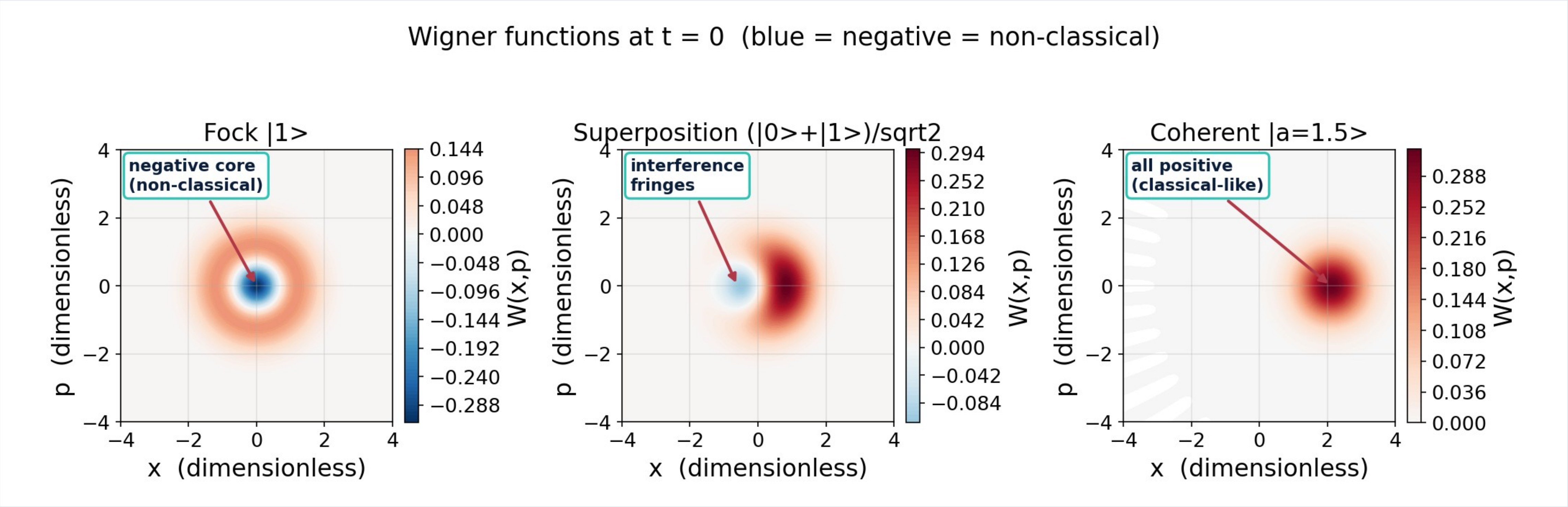


WHAT IT SHOWS

- Eigenvalue error stays at machine-zero for low levels, then blows up.
- Roughly the lowest N/2 levels are exact for a given truncation N.
- Raising N pushes the trustworthy range higher (N=10→5, 30→15, 50→25).
- Rule: increase N until the quantity of interest stops changing.

Component 2 · Task 2

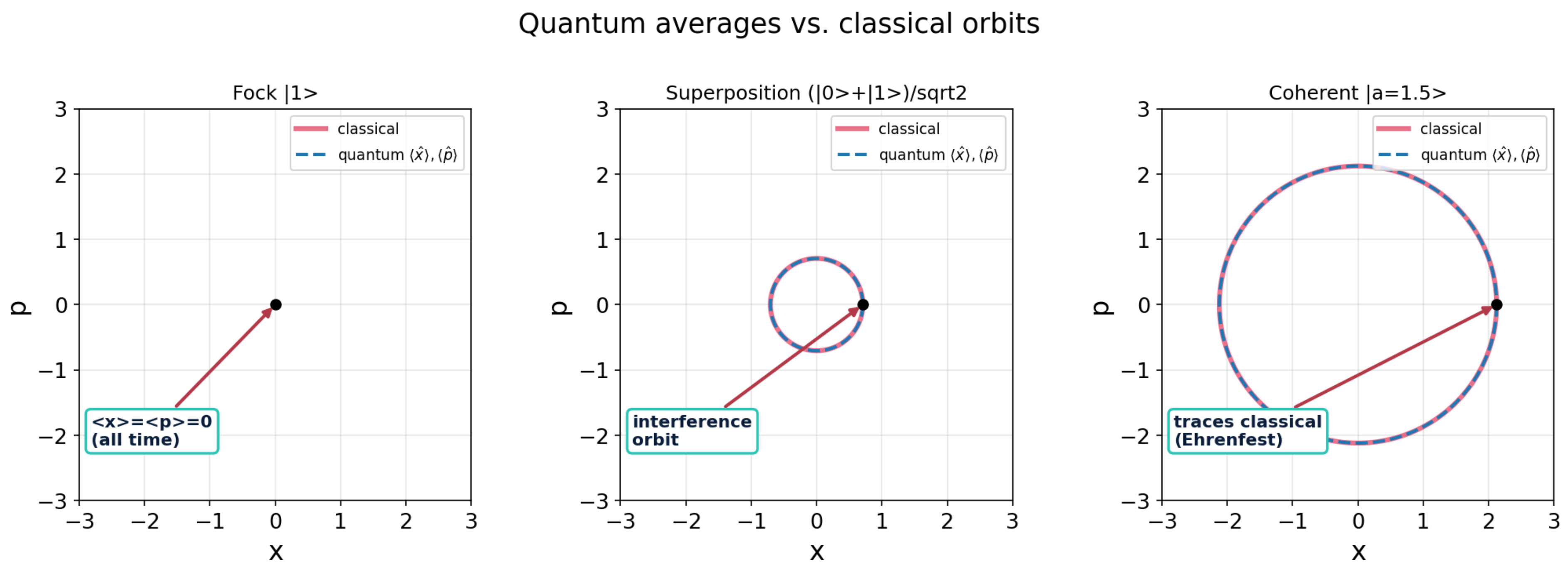
Wigner functions expose non-classicality through negativity and interference.



Takeaway: The Fock state has a negative core and the superposition shows interference fringes (both impossible classically); the coherent state stays positive — the most classical state.

Component 2 · Task 2

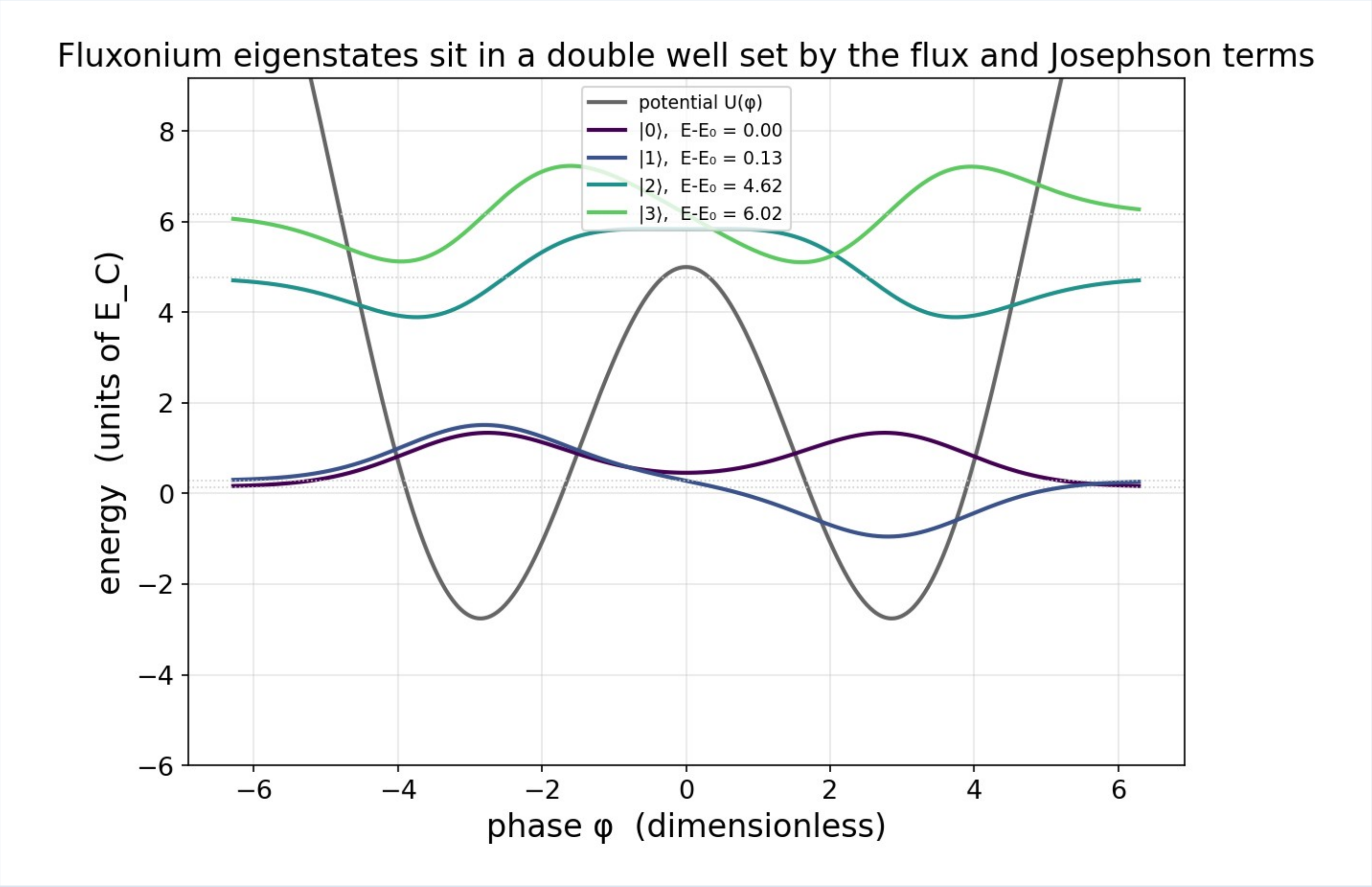
Quantum averages trace the classical orbit, but averaging hides the spread.



Takeaway: For the coherent state ⟨x̂⟩,⟨p̂⟩ follow the classical circle exactly (Ehrenfest's theorem); the Fock state sits at the origin. The averages discard the variance and interference the Wigner picture keeps.

Component 2 · Task 3

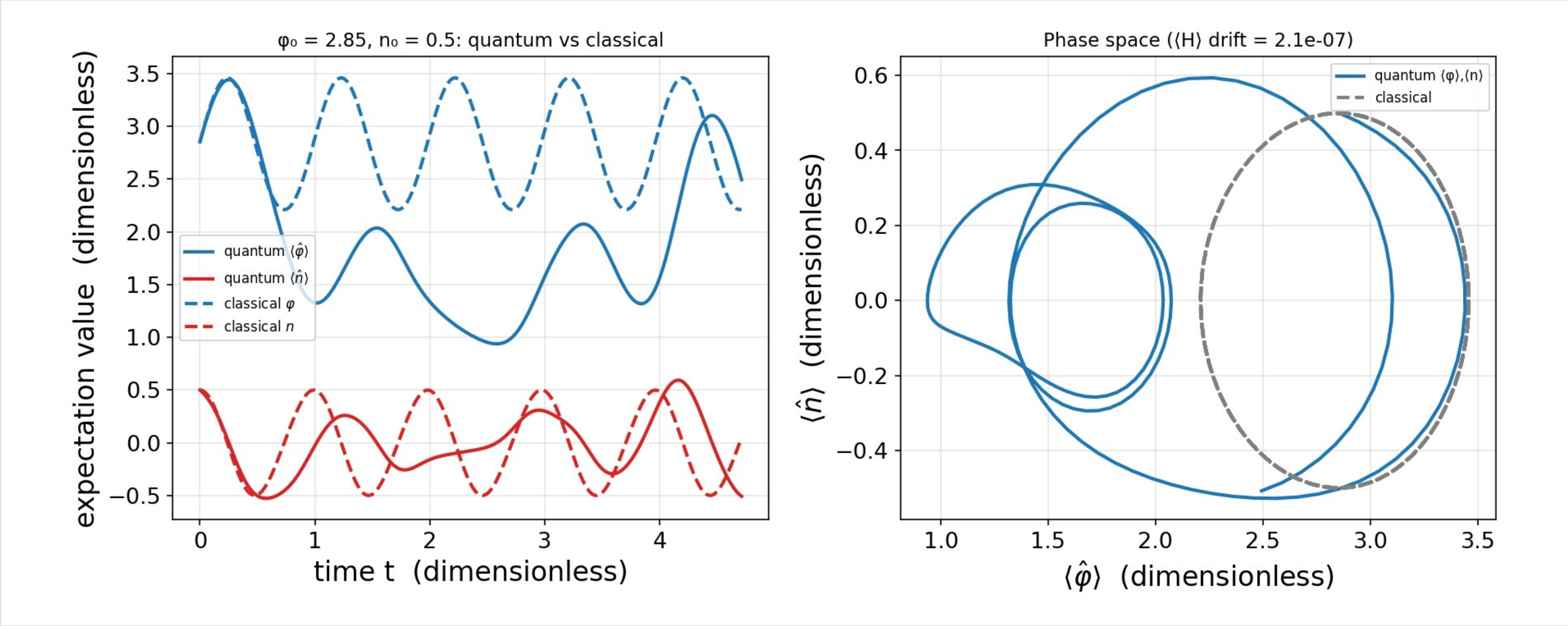
At half flux the fluxonium is a double well whose two lowest states form a tunneling doublet.



WHAT IT SHOWS

- $H = 4E_C \hat{n}^2 + \frac{1}{2}E_L \hat{\varphi}^2 - E_J \cos(\hat{\varphi} + \varphi_{ext})$ — scqubits' convention, at $E_J/E_C = 5$, $E_L/E_C = 0.5$.
- $\varphi_{ext} = \pi$ is the symmetric sweet spot: wells at $\varphi = \pm 2.85$ with a barrier at $\varphi = 0$, 7.76 E_C above them.
- $E_1 - E_0 = 0.134\ E_C$ while $E_2 - E_0 = 4.62\ E_C$ — a 34× gap that isolates a clean two-level system.
- That tiny splitting is tunneling through the barrier, and it has no classical counterpart.

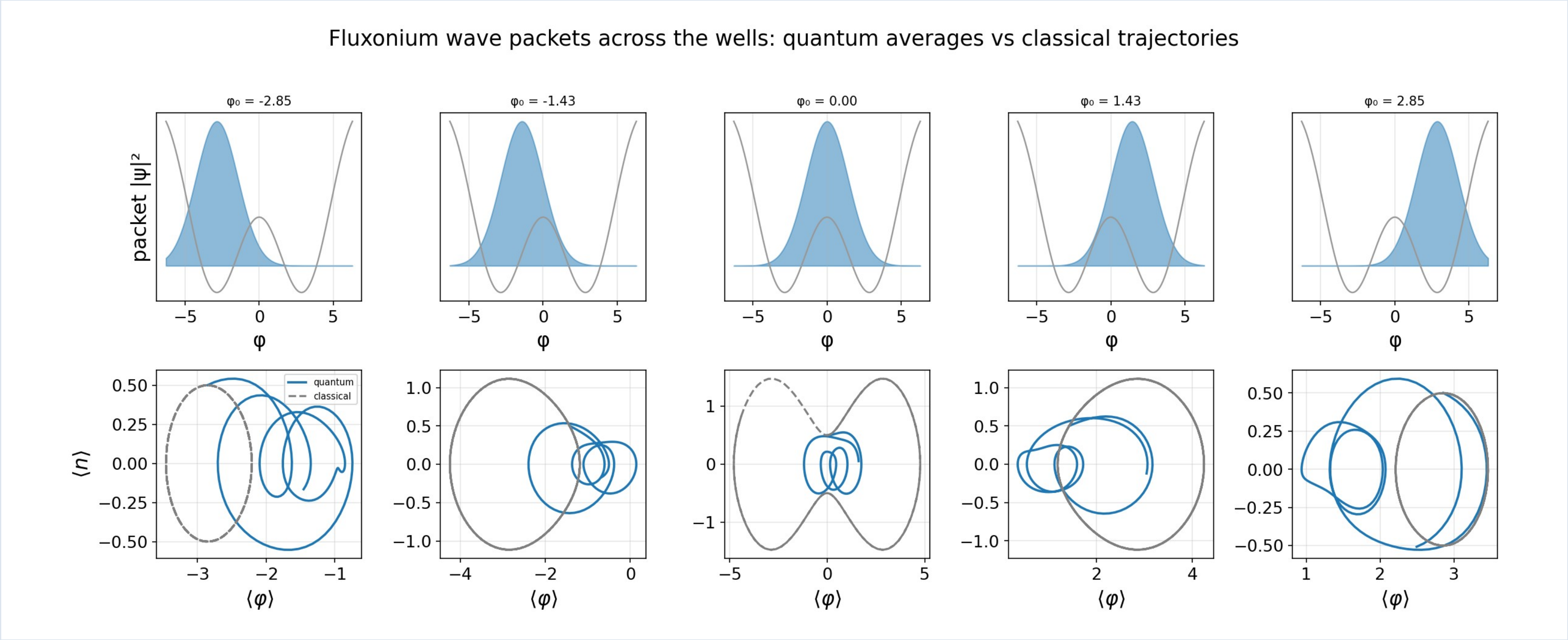
A wave packet tracks the classical orbit for half a period, then falls behind it.



Takeaway: Both start at the well minimum $\varphi_0 = 2.85$ with the same charge kick $n_0 = 0.5$ (placed there at rest, the classical particle sits at an equilibrium and never moves). They agree closely until $t \approx 0.7$, then the quantum average loses amplitude: the packet is as wide as the well, so different parts of it feel different restoring forces. $\langle H \rangle$ is conserved to 2.1e-7, so this is physics, not solver error.

Component 2 · Task 3

Launched on the barrier, the classical and quantum pictures stop describing the same system.



Takeaway: Five starting phases from one well, across the barrier, to the other. In the wells ($\varphi_0 = \pm 2.85$) the two come closest — the orbits share the same region, though the quantum one is still visibly wider. On the barrier top ($\varphi_0 = 0$) the classical point sweeps a wide figure-eight through both wells while the packet splits and stays put — the same initial condition, two completely different answers. That gap is what Component 3 has to learn.

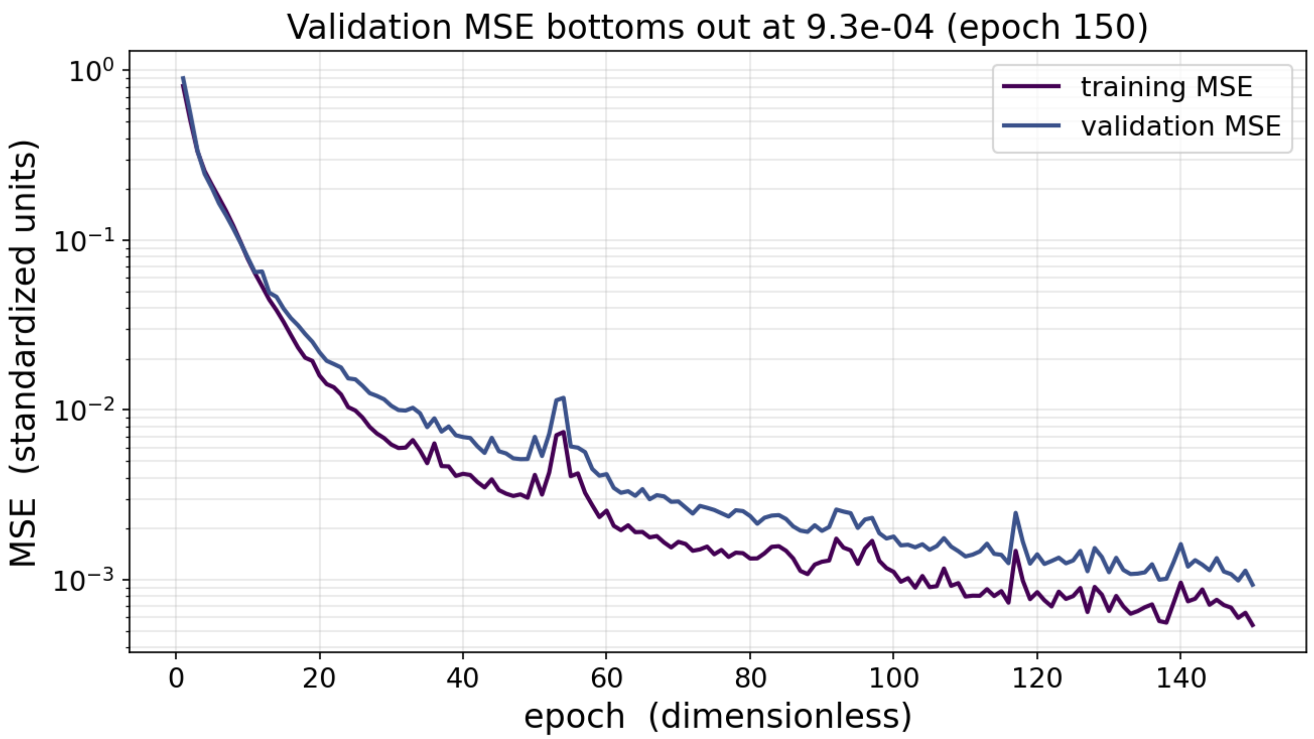
NEW THIS UPDATE

Component 3

The full pipeline runs end to end: paired trajectories → standardize → MLP → validation.

Component 3 · Task 1

Fixing the coordinate convention improved validation MSE nine-fold, to 9.3×10^{-4}.

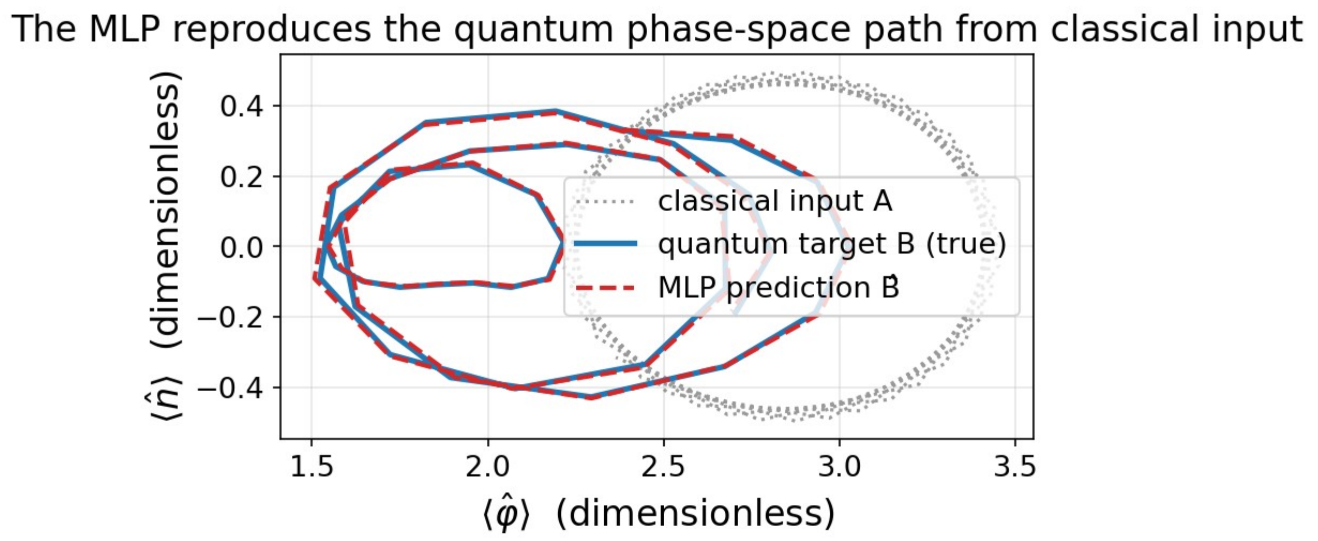


WHAT IT SHOWS

- 300 paired trajectories: classical solve_ivp input A and fluxonium sesolve target B from the same packet.
- Two hidden layers of 128 with ReLU, MSE loss, Adam at lr = 1e-3, an 80/20 split, 150 epochs.
- Validation MSE 9.3e-4 against training 5.4e-4 — a gap of 1.7×, down from 13.7× before the fix.
- Validation was still falling at epoch 150, so the model is under-trained: more epochs should help.

Component 3 · Task 1

On held-out data the predicted quantum path follows the true one, not the classical input.



WHAT IT SHOWS

- One validation trajectory, un-standardized back into physical units.
- Grey dotted is the classical input A; blue is the true quantum target B; red dashed is the prediction.
- The prediction sits on the quantum curve, so the network learned the correction — not a copy of its input.
- Physically consistent pairs are what make this learnable — the same network on mismatched pairs did 9× worse.

Component 3 · Finding

The classical trajectory degrades steadily as the packet starts further up the well.

0.84	0.97	1.19
NEAR THE MINIMUM	MID-WELL	HIGH ON THE WALL
$\lvert\varphi_0 - \varphi\_min\rvert < 0.33$	$0.33 \le \lvert\varphi_0 - \varphi\_min\rvert < 0.66$	$0.66 \le \lvert\varphi_0 - \varphi\_min\rvert < 1.0$

Takeaway: Median RMS difference in φ (radians) between the classical input and the quantum target, over all 300 pairs. The gap never closes: at E_J/E_C = 5 the packet width φ_zpf = 1.41 is comparable to the well itself, so a coherent state is not a classical point particle anywhere in this regime — and it gets worse up the wall.

Verification

Checking every number against an exact formula is what caught the errors.

5×10^{-15}

SPECTRUM

max error vs $E_n = \hbar\omega(n+½)$ over the lowest 15 of N = 30 levels

1.3×10^{-8}

ENERGY DRIFT

over a full period, across all 12 saved classical trajectories

6.7×10^{-9}

SOLVER

max deviation between solve_ivp and the exact analytic solution

Takeaway: The same audit caught the fluxonium work mixing two coordinate conventions — scqubits centres the potential on $\varphi = 0$ (wells at ±2.85, barrier at 0) while the classical side used a copy shifted by φ_ext. Correcting it and re-running improved the Component 3 validation MSE by a factor of nine.

Open questions & next steps

What I don't yet understand

- The classical/quantum gap never drops below ~0.8 rad. How much of that is the packet's width rather than the dynamics?
- Validation was still falling at epoch 150, so the model is under-trained. How far does the MSE go with more epochs?
- Does a deeper well (larger E_J/E_C) bring classical and quantum back together, as the correspondence principle says?
- Should the target be trajectories at all, or a quantum quantity with no classical analogue (tunneling splitting)?

What I'd like to do next

- Extend training past 150 epochs and add early stopping on the validation minimum.
- Widen the sampling window from inside the well out toward the barrier and watch where the MLP starts to fail.
- Add early stopping on the validation minimum, and report component-wise error in physical units.
- Discuss with PI: which fluxonium property is most worth predicting for real device design?